05-14 第1篇

阅读理解（说明文）

① After a rainstorm, you might look up at the sky and see a beautiful rainbow (彩虹). Rainbows have amazed people for thousands of years, but have you ever wondered how they are formed?

② A rainbow appears when sunlight passes through tiny water droplets (水滴) in the air. Sunlight looks white, but it is actually made up of seven colors: red, orange, yellow, green, blue, indigo (靛蓝), and violet (紫罗兰色). When a ray of sunlight enters a water droplet, the light slows down and bends — this is called refraction (折射). Inside the droplet, the light bounces off the back wall like a mirror. Then, as it leaves the droplet, it bends again. Because each color bends at a slightly different angle (角度), the white light is separated into the seven colors we see in a rainbow.

③ To see a rainbow, three conditions must be met. First, there must be water droplets in the air — usually right after rain. Second, the Sun must be shining behind you. Third, the Sun should be fairly low in the sky, less than 42 degrees above the horizon (地平线). This is why rainbows are most common in the early morning or late afternoon, when the Sun is not too high.

④ Sometimes, if you are lucky, you can see a double rainbow (双彩虹). The second rainbow appears above the first one, and its colors are in the opposite order — red is on the inside and violet is on the outside. This happens when sunlight bounces twice inside the water droplets before leaving. The second rainbow is always fainter (更淡的) than the first because some light is lost with each bounce.

⑤ Rainbows are not just beautiful — they have also inspired (启发) people throughout history. In many cultures, a rainbow is a symbol (象征) of hope and good luck. Scientists like Isaac Newton studied rainbows carefully and helped us understand the nature of light. Next time you see a rainbow, remember: you are watching science and beauty come together in the sky.

注释： rainbow 彩虹；droplet 水滴；refraction 折射；angle 角度；horizon 地平线；faint 微弱的；inspire 启发；symbol 象征

选择题

1. According to paragraph ②, what happens when sunlight enters a water droplet? A. The light becomes hotter and stronger. B. The light slows down and bends. C. The light disappears completely. D. The light turns into water.

2. How many colors make up white sunlight? A. Three. B. Five. C. Seven. D. Ten.

3. Which of the following is NOT a condition for seeing a rainbow? A. There must be water droplets in the air. B. The Sun must be shining behind you. C. The temperature must be below zero. D. The Sun should be fairly low in the sky.

4. What can we learn about double rainbows from paragraph ④? A. The second rainbow has brighter colors than the first. B. The second rainbow has colors in the opposite order. C. Double rainbows only appear at night. D. Double rainbows are caused by strong wind.

5. What is the main purpose of this passage? A. To tell a story about a child who chased a rainbow. B. To explain how rainbows are formed and describe their features. C. To persuade people to study science at university. D. To compare rainbows with other weather events.

详细解析

1. 【答案】B

【题型】 细节理解

【定位】 第②段第4句 "When a ray of sunlight enters a water droplet, the light slows down and bends — this is called refraction."

【解析】 原文明确说明阳光进入水滴后会减速并弯曲，这称为折射。A 项"光变得更热更强"文中未提及；C 项"光完全消失"与形成彩虹的事实矛盾；D 项"光变成水"属于无中生有。故选 B。

2. 【答案】C

【题型】 细节理解

【定位】 第②段第2句 "Sunlight looks white, but it is actually made up of seven colors: red, orange, yellow, green, blue, indigo, and violet."

【解析】 原文列出白色阳光由七种颜色组成，并逐一列出。A 项"三种"、B 项"五种"和 D 项"十种"均与原文数字不符。故选 C。

3. 【答案】C

【题型】 细节理解

【定位】 第③段列出的三个条件：空气中有水滴、太阳在身后照耀、太阳位置较低。

【解析】 原文列举了看到彩虹的三个条件，其中没有提到"温度必须低于零度"。A、B、D 三项均是原文明确提到的条件。C 项"温度必须低于零度"属于无中生有。故选 C。

4. 【答案】B

【题型】 细节理解

【定位】 第④段 "The second rainbow appears above the first one, and its colors are in the opposite order — red is on the inside and violet is on the outside."

【解析】 原文说第二条彩虹的颜色顺序与第一条相反。A 项"颜色更亮"与"always fainter"矛盾；C 项"只在夜间出现"属于无中生有，彩虹需要阳光；D 项"由强风造成"文中未提及。故选 B。

5. 【答案】B

【题型】 写作意图

【定位】 全文：第①段引出话题，第②段解释形成原理，第③段说明观察条件，第④段介绍双彩虹，第⑤段总结意义。

【解析】 全文以科普方式解释彩虹的形成原理、观察条件及特征，属于说明文。B 项准确概括了文章目的。A 项"讲述追彩虹的故事"属于记叙文；C 项"说服人们学科学"过度推断；D 项"比较彩虹和其他天气现象"文中未涉及比较。故选 B。